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Games of Complete Information

1.1 Static Games of Complete Information

We start with a classic story:

Prisoners' Dilemma

Two suspects are arrested and charged with a crime. The police lack sufficient evidence to convict the suspects, unless at least one confesses. The suspects are held in separate cells and told:

- If only one of you confesses and testifies against your partner, then the person who confesses will go free while the person who does not confess will be convicted and given a 20-year jail sentence.
- If both of you confess, then both will be convicted and sent to prison for 5 years.
- Finally, if neither of you confesses, then both of you will be convicted of a minor offence and sentenced to 1 year in jail.

What should the suspects do?

To investigate the situation, we can represent the prisoners' decisions using a matrix:

	Holdout	Confess
Holdout	$(-1, -1)$	$(-20, 0)$
Confess	$(0, -20)$	$(-5, -5)$

With some observation, it is easy to see that no matter what the other prisoner chooses to do, the best decision for the prisoner is to confess. Note that in this representation, we represent a *strategy* with an ordered pair consisting of the decisions made by the two prisoners respectively. Extending this idea to a higher dimension, we see that every strategy of a game is formed by an ordered n -tuple, where each factor represents a decision taken by the corresponding player.

Definition 1.1.1 ► Strategy Space

In an n -player game, the **strategy space** of the i -th player, denoted by S_i , is the set of all possible decisions for the player.

In the above example, for each strategy (i, j) , we then place the years of imprisonment, or the *payoff*, for the two prisoners following that strategy in the corresponding matrix entry. Intuitively, for an n -player game in general, we can assign a payoff value for each strategy in $\prod_{i=1}^n S_i$, the set of all strategies.

Definition 1.1.2 ► Payoff Function

Let S_i be the strategy space of the i -th player in an n -player game, the **payoff function** for the i -th player is a map $u_i : \prod_{j=1}^n S_j \rightarrow \mathbb{R}$.

This gives us a convenient way to represent a game quantitatively.

Definition 1.1.3 ► Normal-Form of Games

Let G be an n -player game. Let S_i be the strategy space and u_i be the payoff function of the i -th player, then the **normal-form**, or **strategic form** of G is defined as

$$G := \{S_1, S_2, \dots, S_n, u_1, u_2, \dots, u_n\}.$$

It is important to note that the payoff a player depends not only on his own action but also on the actions of others. This inter-dependence is the essence of games.

Remark. For 2-player games, the payoff can be represented in a matrix.

Given any game, our goal is to derive the optimal strategy for the players. Before that, we shall discuss some criteria under which a strategy cannot be optimal. For the sake of notational simplicity, we denote

$$\mathbf{s}_{-i} := (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$$

and

$$\mathbf{s} := (s_1, s_2, \dots, s_n) = (s_i, \mathbf{s}_{-i}).$$

Analogously, we denote

$$S_{-i} := \prod_{j=1}^{i-1} S_j \times \prod_{j=i+1}^n S_j.$$

Definition 1.1.4 ► Dominated and Dominant Strategies

Let $G := \{S_1, S_2, \dots, S_n, u_1, u_2, \dots, u_n\}$ be a normal-form game and $s_i, s'_i \in S_i$ be two strategies for the i -th player. We say that s_i is **strictly dominated** by s'_i or that s'_i is

strictly dominant over s_i if

$$u_i(s_i, \mathbf{s}_{-i}) < u_i(s'_i, \mathbf{s}_{-i}) \quad \text{for all } \mathbf{s}_{-i} \in S_{-i}.$$

Remark. In other words, s_i is strictly dominated by s'_i if and only if it is a strictly worse decision no matter what the other players decide to do.

If a strategy s is strictly dominated by any other strategy, then it is not a rational strategy because it can never be optimal.

Assuming that in an n -player game, every player knows exactly what the strategies are and their respective payoffs, then we can proceed with the following reasoning:

- Suppose that s_{i_0} is strictly dominated by any other strategy of player i , then if the player is rational, the strategy will never be played.
- Therefore, we can reduce the strategy space to $\{\mathbf{s} \in \prod_{i=1}^n S_i : s_i \neq s_{i_0}\}$.
- Apply the same reasoning on the reduced strategy space iteratively. If at each iteration there exists such a strictly dominated strategy, then we can successfully obtain the optimal strategy for the n -players.

However, in most games, we might not be able to find such a strictly dominated strategy. Therefore, we have to think from a conditional context: given a set of strategies played by the other players, what is the *best response* for the current player?

Definition 1.1.5 ► Best-Response Correspondence

Let $G := \{S_1, S_2, \dots, S_n, u_1, u_2, \dots, u_n\}$ be a normal-form game. The **best-response correspondence** for player i is a map $R_i : S_{-i} \rightarrow \mathcal{P}(S_i)$ defined by

$$R_i(\mathbf{s}_{-i}) := \operatorname{argmax}_{s_i \in S_i} u_i(s_i, \mathbf{s}_{-i}).$$

Now, given $\mathbf{s}_{-i}$, the actions played by the other players, $R_i(\mathbf{s}_{-i})$ tells us what the best responses are for player i for his own profit. In other words, player i has **no incentive** to choose any strategy in $S_i \setminus R_i(\mathbf{s}_{-i})$. If such a situation holds for all the players, then it is clear that the game has reached an *equilibrium* where no player is incentivised to deviate from the current strategy.

Definition 1.1.6 ▶ Nash Equilibrium

Let $G := \{S_1, S_2, \dots, S_n, u_1, u_2, \dots, u_n\}$ be a normal-form n -player game, then the strategies $(s_1^*, s_2^*, \dots, s_n^*)$ are a **Nash equilibrium** if

$$s_i^* \in R_i(s_{-i}^*) \quad \text{for all } i = 1, 2, \dots, n,$$

or equivalently,

$$u_i(s_i^*, s_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, s_{-i}^*) \quad \text{for all } i = 1, 2, \dots, n.$$